STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-PULLBACK-INJECTIVE-MODULES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3043-3064 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 3066-3087, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_algebraic-stacks-sheaves.json (pb=945, stmt 3043-3064, proof 3066-3087).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 015Z: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3087–3087. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-localizing-structure-sheaf: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1345–1354. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-pullback-injective: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3009–3022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0797: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3087–3087. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0798: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 3087–3087. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. Let $\tau \in \{Zar, étale, smooth, syntomic, fppf\}$. The functor $f^* : Mod(\mathcal{Y}_\tau, \mathcal{O}_\mathcal{Y}) \to Mod(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X})$ has a left adjoint $f_! : Mod(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X}) \to Mod(\mathcal{Y}_\tau, \mathcal{O}_\mathcal{Y})$ which agrees with the functor $f_!$ of Lemma lemma-pullback-injective on underlying abelian sheaves. If f is faithful and $\mathcal{X}$ has equalizers, then*

(1) *$f_!$ is exact, and*
(2) *$f^{-1}\mathcal{I}$ is injective in $Mod(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X})$ for $\mathcal{I}$ injective in $Mod(\mathcal{Y}_\tau, \mathcal{O}_\mathcal{X})$.*

Proof. Recall that f is a continuous and cocontinuous functor of sites and that $f^{-1}\mathcal{O}_\mathcal{Y} = \mathcal{O}_\mathcal{X}$. Hence Modules on Sites, Lemma 0797 implies f^* has a left adjoint $f_!^{Mod}$. Let x be an object of $\mathcal{X}$ lying over the scheme U. Then f induces an equivalence of ringed sites

$$\mathcal{X}/x \longrightarrow \mathcal{Y}/f(x)$$

as both sides are equivalent to $(Sch/U)_\tau$, see Lemma lemma-localizing-structure-sheaf. Modules on Sites, Remark 0798 shows that $f_!$ agrees with the functor on abelian sheaves.

Assume now that $\mathcal{X}$ has equalizers and that f is faithful. Lemma lemma-pullback-injective tells us that $f_!$ is exact. Finally, Homology, Lemma 015Z implies the statement on pullbacks of injective modules. □